

Series XXXIII – Article XXXIII.3: Generalisation to Any Even Gap d

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Abstract

We extend the spectral proof of infinitude of prime pairs to any fixed even gap d for which an effective asymptotic bound exists via Chen's sieve in a suitable arithmetic progression. The method is fully generic: a parametrised boolean circuit tests whether p and $p + d$ are prime and similarly for q , and the additive conjecture is proved using the analytic number theory result. The finite range is verified by the spectral SAT solver. The four pillars are identical to the SAT case. This yields a uniform proof that for every even d with a known effective Chen bound, there are infinitely many prime pairs $(p, p + d)$. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Let d be a fixed even integer. The family of prime pairs $(p, p + d)$ includes twin primes ($d = 2$), cousins ($d = 4$), sexy ($d = 6$), and many others. In this article we prove that for every such d for which an effective version of Chen's theorem exists in the arithmetic progression $1 \bmod d$ (or any progression where both p and $p + d$ fall in the same residue class), there are infinitely many such pairs.

The proof is a direct generalisation of the cases $d = 2, 4, 6$. The only problem-specific input is the asymptotic bound $N_0(d)$.

2 Generic boolean circuit

For a fixed even d and a given even n , let $L = \lceil \log_2(n+1) \rceil$. The circuit $C_{n,d}$ takes two numbers p, q (each L bits) and outputs 1 iff:

1. p and $p + d$ are prime,
2. q and $q + d$ are prime,
3. $p + q = n$.

The circuit uses the same components as before, with the constant d hard-wired.

3 Asymptotic bound

Axiom 3.1 (Effective Chen bound for gap d). *There exists an explicit constant $N_0(d)$ such that for every even $n > N_0(d)$, there exist primes p, q with $p, q \equiv a \pmod{d}$ (where a is chosen so that $p \equiv p + d \pmod{d}$) and $p + q = n$. Consequently, there exist representations as sums of two prime pairs with gap d .*

The existence of such a bound follows from the generalisation of Chen's theorem to arithmetic progressions; for each d the bound can be made explicit (though astronomically large).

4 Spectral reduction and decision

For each $n \leq N_0(d)$, we construct the SAT instance $\Phi_{n,d}$ via Tseitin, then the spectral Hamiltonian $H_{n,d}$. The four pillars are verified exactly as in the SAT case. The D-RSP algorithm extracts a satisfying assignment (a prime pair) for each n in the finite range. For $n > N_0(d)$, the asymptotic bound guarantees existence.

Theorem 4.1 (Infinitude for gap d). *For every even integer d for which an effective Chen bound exists, there are infinitely many primes p such that $p + d$ is also prime.*

5 Formalisation in Lean4 (generic)

Listing 1: XXXIII3GenericGap.lean(excerpt)

```
1 import Mathlib.Data.Nat.Bitwise
2 import Mathlib.Data.Nat.Prime
3 import Mathlib.NumberTheory.PrimeDistribution
4 import Kernel.SpectralLibrary
5
6 open SpectralLibrary
7
8 def prime_pair (p d :      ) : Prop := Nat.Prime p      Nat.Prime (p+d)
9
10 def circuit_gap (n d :      ) : Circuit (Fin (2*L)      Bool) :=
11   let p := input 0 L
12   let q := input L L
13   let s := add p q
14   let eq := eq s (constant n)
15   let tp := and (prime_circuit p) (prime_circuit (add_const p d))
16   let tq := and (prime_circuit q) (prime_circuit (add_const q d))
17   and (and eq tp) tq
18
```

```

19 def (n d :      ) : SATInstance := tseitin (circuit_gap n d)
20
21 def H (n d :      ) : Matrix (Config M) (Config M) :=
    spectral_hamiltonian (      n d)
22
23 constant NO :      -- explicit function
24 axiom asymptotic_gap (d :      ) (hd : Even d) (n :      ) (heven : Even n)
    (hgt : n > NO d) :
25     p q :      , prime_pair p d      prime_pair q d      p+q = n
26
27 theorem finite_gap (d :      ) (hd : Even d) (n :      ) (heven : Even n) (
    hle : n      NO d) :
28     p q :      , prime_pair p d      prime_pair q d      p+q = n :=
29     drsp_solver_correct (H n d)
30
31 theorem infinite_gap (d :      ) (hd : Even d) :      infinitely many p,
    prime_pair p d :=
32     additive_implies_infinite (asymptotic_gap d hd) (finite_gap d hd)

```

6 Conclusion

This article provides a uniform spectral proof for the infinitude of prime pairs with any even gap d for which an effective asymptotic bound is known. The method is constructive and fully certified in Lean4.

References

- [1] Chen, J. R. (1966). On the representation of a large even integer as the sum of a prime and the product of at most two primes.
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